CSS Media Queries allow you to apply different styles based on the screen's size, resolution, orientation, or device type. They're essential for building **responsive websites**.

🔹 Basic Media Query Syntax

```
@media media-type and (condition) {
  /* CSS rules here */
}
```

Now, let's look at **common and practical examples**:

✅ Example 1: Mobile First (Adjust Font Size)

```
body {
  font-size: 16px;
}

@media (min-width: 768px) {
  body {
    font-size: 18px;
  }
}

@media (min-width: 1024px) {
  body {
    font-size: 20px;
  }
}
```

🟢 **Explanation:** Font size increases with screen width (mobile → tablet → desktop).

✅ Example 2: Hide Sidebar on Small Screens

```
css
CopyEdit
.sidebar {
  display: block;
}

@media (max-width: 600px) {
  .sidebar {
    display: none;
  }
```

```
}
```

🟢 **Explanation:** Hides the `.sidebar` on screens **600px or smaller**.

✅ Example 3: Change Layout Using Flexbox

```
.container {
  display: flex;
  flex-direction: row;
}

@media (max-width: 768px) {
  .container {
    flex-direction: column;
  }
}
```

🟢 **Explanation:** Switches layout from **horizontal** to **vertical** on tablets and phones.

✅ Example 4: Responsive Image Size

```
img.responsive {
  width: 100%;
  height: auto;
  max-width: 400px;
}

@media (max-width: 500px) {
  img.responsive {
    max-width: 100%;
  }
}
```

🟢 **Explanation:** Image scales down for smaller screens but caps at `400px` for larger ones.

✅ Example 5: Target Portrait vs Landscape

```
@media (orientation: portrait) {
  body {
    background-color: lightblue;
  }
}
```

```
@media (orientation: landscape) {
  body {
    background-color: lightgreen;
  }
}
```

🟢 **Explanation:** Changes background color based on device orientation.

✅ Example 6: Dark Mode Support

```
@media (prefers-color-scheme: dark) {
  body {
    background-color: #121212;
    color: white;
  }
}
```

🟢 **Explanation:** Applies dark theme when the user's system prefers it.

✅ Example 7: Responsive Grid

```
.grid {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
}

@media (max-width: 1024px) {
  .grid {
    grid-template-columns: repeat(2, 1fr);
  }
}

@media (max-width: 600px) {
  .grid {
    grid-template-columns: 1fr;
  }
}
```

🟢 **Explanation:** Adjusts the number of columns depending on the screen size.